MILITARY SPECIFICATION SHEET
GUIDANCE CONTROL SECTION, GUIDED MISSILE
POWER SUPPLY ASSEMBLY

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the power supply assembly described herein shall consist of this specification and the latest issue of MIL-G-85742.

1. SCOPE

1.1 Scope. This specification sheet establishes the performance and test requirements for the power supply assembly used in the Guidance Control Section of the SIDEARM Missile.

2. APPLICABLE DOCUMENTS.

2.1 Government drawings. The following drawings, in addition to the applicable documents listed in MIL-G-85742, are applicable to this specification sheet.

Naval Air Systems Command
(Code Ident 30003)

1466AS1302 Power Supply Assembly.

3. REQUIRMENTS

3.1 Item description. The power supply assembly is a part of the receiver section. It provides regulated ±27 volts (V), ±15 V, and +5 V supplies as well as a +15 V supply with a floating ground for the yttrium iron garnet (YIG) oscillator in the receiver subassembly. The power supply also provides unregulated ± 35 V supplies and an 86 volts root-mean-square (Vrms) supply to the seeker section. The gyro drive board of the power supply generates a three-phase voltage to drive the head gyro in the seeker section.

FSC 1420

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3.2 Characteristics.

3.2.1 Performance. Performance characteristics of the power supply shall be as specified herein.

3.2.1.1 Input power. The power supply shall operate on an input voltage of 138 + 7 Vrms at a frequency of 8.0 ± 0.3.kilohertz (kHz).

3.2.1.2 Output power. The output voltages shall meet the requirements of Table I.

Table I. Output voltages.

Voltage	Pin #	Return	Tolerance	Load	Ripple volts peak-to-peak	Frequency
+15 V	7	18	±0.15 V	750 mA	0.100	dc
+15 V INTL PS	23	18	±0.15 V	---	---	dc
-15 V	20	18	±0.15 V	375 mA	0.100	dc
+27 V	21	25	±1 V	53 mA	0.100	dc
-27 V	8	25	±1 V	85 mA	0.100	dc
+ 5 V	10	18	±0.15 V	667 mA	0.100	dc
+15 V LO	9	22	±0.30 V	1 A	0.100	dc
+35 V	6	12	±4 V	280 mA	1	dc
-35 V	19	12	±4 V	280 mA	1	dc
86 VAC	4	17	±10 Vrms	287 mA	---	8 ±0.3 kHz

3.2.1.3 Gyro drive voltage. With a seeker section as the load, the three-phase gyro drive voltage shall be 19.0 ± 1.5 Vrms.

3.2.1.4 Gyro frequency. The voltages of 3.2.1.3 shall have a frequency and phasing such that the seeker gyro spins in a clockwise direction at a steady state speed of F1 as specified in the Appendix.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General. Each power supply shall meet the requirements of 3.2 when inspect-tested as specified in accordance with Drawing 1466AS1302 and the following paragraphs. Failure to meet any of the requirements shall be cause for rejection of the power supply.

4.1.1 Test environment. Unless otherwise specified, all tests and inspections shall be conducted under the following environmental conditions:

a. Temperature (ambient): 23 ± 5 degrees Celsius.

b. Altitude: Elevation of test facility.

c. Relative humidity: 95 percent or less.

4.1.2 Test equipment accuracy. Unless otherwise specified, test equipment accuracy shall be as specified in MIL-G-85742.

4.2 Quality conformance inspections.

4.2.1 Unit test. The assembly shall be subjected to a unit test in order to establish conformance to the requirements specified herein.

4.2.2 Input voltage. The voltage applied at pins 3 and 15 with return at pin 16 of A2A3P1, shall meet the requirements of 3.2.1.1.

4.2.3 Test methods. Details of methods and procedures shall be in accordance with the test procedure.

4.2.3.1 Output power. Measure the output voltages at connector A2A3P1 using the pin numbers given in Table I. The voltages shall meet the requirements of Table I.

4.2.3.2 Gyro drive voltages. With the seeker section as a load, measure the three-phase gyro drive voltages on pin 1 (phase A), pin 2 (phase B), and pin 14 (phase C) of the connector, A2A3P1 using pin 18 for return. The voltages shall meet the requirements of 3.2.1.3.

4.2.3.3 Gyro frequency. Measure the gyro frequency on pin 20 of the seeker load (connector A1P1). The frequency shall meet the requirments of 3.2.1.4.

Preparing Activity
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(Project 1420-NO3)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-85742/3(AS)	Power Supply Assembly

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Includ Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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